

LECTURE 4 – PROBABILITY AND OPTIMISATION

INTRODUCTION TO AUTONOMOUS SYSTEMS · MODULE II

Topics: Probability · Bayesian inference · Entropy · Convex optimisation · MAP estimation

AGENDA

S	TOPIC	KEY IDEAS
1	Why Two Languages?	Inference vs optimisation · algorithm taxonomy
2	Probability Recap	Probability spaces · Gaussian · entropy · KL
3	Bayes' Theorem	Prior · likelihood · posterior · belief update
4	Optimisation Recap	Convexity · gradients · KKT conditions
5	MAP Estimation	Inference as optimisation · Tikhonov regularisation
6	The Duality	Every algorithm is inference, optimisation, or both

LITERATURE

TOPIC	REFERENCE
Probability theory	Гнеденко — <i>Курс теорії імовірностей</i> (укр. вид.); Billingsley — <i>Probability and Measure</i> (3rd ed., 1995)
Bayesian inference and ML	Murphy — <i>Probabilistic Machine Learning: An Introduction</i> (2022); Bishop — <i>PRML</i> (2006)
Information theory	Cover & Thomas — <i>Elements of Information Theory</i> (2nd ed., 2006)
Convex analysis and optimisation	Boyd & Vandenberghe — <i>Convex Optimization</i> (2004); Nocedal & Wright — <i>Numerical Optimization</i> (2nd ed., 2006)

S1 · WHY TWO LANGUAGES?

EVERY ALGORITHM IN THIS COURSE IS INFERENCE, OPTIMISATION, OR BOTH



TWO PROBLEMS FROM ONE TUPLE

The tuple $(\mathcal{X}, \mathcal{O}, \mathcal{A}, f, h, \pi)$ produces two distinct computational problems:

Inference. Given observations $o_{1:t}$ and models f, h , what is the hidden state x_t ?

$$b_t = p(x_t \mid o_{1:t}) \quad (\text{the belief — a distribution})$$

Optimisation. Given a model of the system, find the best policy:

$$\pi^* = \arg \min_{\pi} \mathcal{L}(\pi) \quad (\text{a scalar objective})$$

Every algorithm in this course is either inference, optimisation, or both — and they are more deeply related than they first appear.

ALGORITHM TAXONOMY

ALGORITHM	LANGUAGE	MODULE
Kalman filter	Inference	III
Particle filter	Inference	III
A*, PID	Optimisation (implicit)	IV
LQR	Optimisation	IV
MPC	Optimisation (repeated)	IV
Behavioural cloning	Optimisation	V
Reinforcement learning	Both — interleaved	V
Diffusion policy	Both — separated	VI

Understanding each language in isolation makes their combination in RL legible.

S2 · PROBABILITY RECAP

THE FORMAL SCAFFOLDING FOR BELIEF



§2.1 · THE PROBABILITY SPACE

Definition (σ -algebra). A σ -algebra on Ω is a collection $\mathcal{F} \subseteq 2^\Omega$ satisfying:

1. $\Omega \in \mathcal{F}$;
2. $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$; (closed under complement)
3. $A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$; (closed under countable unions)

Elements of $\mathcal{F}$ are called **measurable sets** or **events**.

Property. Any σ -algebra is also closed under countable intersections: $\bigcap_i A_i = (\bigcup_i A_i^c)^c \in \mathcal{F}$.

On finite Ω : take $\mathcal{F} = 2^\Omega$ (all subsets) — every subset can be consistently assigned a probability, so there is no reason to exclude any event.

On $\mathbb{R}$: $2^\mathbb{R}$ fails (non-measurable sets exist); use the **Borel σ -algebra** $\mathcal{B}(\mathbb{R}) = \sigma(\text{open intervals})$ — the **smallest** σ -algebra on $\mathbb{R}$ containing all open intervals. It contains every event arising in analysis and probability, while excluding non-measurable sets.

PROBABILITY MEASURE AND PROBABILITY SPACE

Definition (Probability measure). A function $P : \mathcal{F} \rightarrow [0, 1]$ satisfying the **Kolmogorov axioms**:

1. **Non-negativity:** $P(A) \geq 0$ for all $A \in \mathcal{F}$
2. **Normalisation:** $P(\Omega) = 1$
3. **σ -additivity:** for pairwise disjoint $A_1, A_2, \dots \in \mathcal{F}$:
$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

From these axioms: $P(\emptyset) = 0$, $P(A^c) = 1 - P(A)$, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Definition (Probability space). A triple $(\Omega, \mathcal{F}, P)$ where Ω is the **sample space**, $\mathcal{F}$ is a σ -algebra on Ω , and P is a probability measure on $(\Omega, \mathcal{F})$.

EXAMPLES OF PROBABILITY SPACES

SETTING	Ω	$\mathcal{F}$	P
Fair coin	$\{H, T\}$	$2^\Omega = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$	$P(\{H\}) = P(\{T\}) = \frac{1}{2}$
Fair die	$\{1, \dots, 6\}$	2^Ω	$P(\{k\}) = \frac{1}{6}$ for each k
Uniform on $[0, 1]$	$[0, 1]$	$\mathcal{B}([0, 1])$	$P((a, b)) = b - a$ (Lebesgue measure)
Robot state	$\mathbb{R}^2 \times [0, 2\pi)$	$\mathcal{B}(\mathbb{R}^2 \times [0, 2\pi))$	$P(A) = \int_A p(x, y, \theta) dx dy d\theta$

An **event** is always a *subset* of Ω — the set of all outcomes where some condition holds. For the robot: "within 1 m of the goal" is $A = \{(x, y, \theta) : \sqrt{(x - x_g)^2 + (y - y_g)^2} \leq 1\}$, and $P(A) = \int_A p dx dy d\theta$.

RANDOM VARIABLES AND DISTRIBUTIONS

$(\Omega, \mathcal{F}, P)$ assigns probabilities to raw outcomes, but Ω is often abstract and unwieldy. We care about a specific numerical quantity extracted from the outcome — a sensor reading, a robot position, the number of heads. A **random variable** is the function that performs this extraction: it maps each $\omega \in \Omega$ to the quantity of interest, letting us do arithmetic in a concrete space $\mathcal{X}$.

Definition (Random variable). A **measurable** function $X : \Omega \rightarrow \mathcal{X}$, where $(\mathcal{X}, \mathcal{G})$ is a measurable space ($\mathcal{G}$ is a σ -algebra on the output space $\mathcal{X}$). Measurability requires: for every output event $B \in \mathcal{G}$,

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$$

so that $P(X \in B) = P(X^{-1}(B))$ is well-defined.

Definition (PMF). When $\mathcal{X}$ is countable: $p(x) = P(X = x) \geq 0$, $\sum_{x \in \mathcal{X}} p(x) = 1$.

Definition (PDF). When $\mathcal{X} = \mathbb{R}^n$: $p(x) \geq 0$, $\int_{\mathbb{R}^n} p(x) dx = 1$, and $P(X \in A) = \int_A p(x) dx$ for any Borel A .

EXAMPLES OF RANDOM VARIABLES

SETTING	Ω	$\mathcal{X}$	$X(\Omega)$	DISTRIBUTION
Door sensor	All experimental runs	$\{0, 1\}$	Sensor output bit	$\text{Ber}(\theta)$
Coin flips	All sequences of 3 flips	$\{0, 1, 2, 3\}$	Number of heads	$\text{Bin}(3, \frac{1}{2})$
Robot position	All physical histories	$\mathbb{R}^2$	(p_x, p_y) at time t	$\mathcal{N}(\mu, \Sigma)$
Sensor range	All experimental runs	$\mathbb{R}_{>0}$	Distance reading	e.g. exponential

Ω provides the randomness; X extracts the quantity of interest. The **law** of X is the induced measure $P_X(B) = P(X^{-1}(B))$ on $\mathcal{X}$ — this is the object we compute with. The underlying Ω recedes into the background.

JOINT, MARGINAL, CONDITIONAL

Definition (Joint distribution). $p(x, y)$ – the law of the pair (X, Y) .

Definition (Marginal distribution). $p(x) = \int p(x, y) dy$; (law of total probability)

Definition (Conditional distribution). $p(x | y) = \frac{p(x, y)}{p(y)}$; for $p(y) > 0$

Proposition (Chain rule). For any random variables X and Y :

$$p(x, y) = p(x | y) p(y) = p(y | x) p(x)$$

Equating the two right-hand sides and solving for $p(x | y)$ gives Bayes' theorem.

EXPECTATION AND COVARIANCE

Definition (Expectation). $\mathbb{E}[X] = \int x p(x) dx$; more generally $\mathbb{E}[g(X)] = \int g(x) p(x) dx$.

Definition (Covariance matrix). $\Sigma = \text{Cov}(X) = \mathbb{E}[(X - \mu)(X - \mu)^\top] \in \mathbb{R}^{n \times n}$, where $\mu = \mathbb{E}[X]$.

Proposition (Linearity of expectation). For any X, Y and constants $a, b \in \mathbb{R}$:

$$\mathbb{E}[aX + bY] = a \mathbb{E}[X] + b \mathbb{E}[Y]$$

This holds regardless of any dependence between X and Y .

Proposition (Law of total expectation). For any random variables X, Y and measurable g :

$$\mathbb{E}[g(X)] = \mathbb{E}_Y[\mathbb{E}[g(X) | Y]]$$

Proof. $\mathbb{E}_Y[\mathbb{E}[g(X)|Y]] = \int \int g(x) p(x|y) dx p(y) dy = \int g(x) \int p(x|y)p(y) dy dx = \mathbb{E}[g(X)]. \square$

PROPERTIES OF COVARIANCE

Proposition (Covariance is symmetric PSD). $\Sigma = \Sigma^\top$ and $v^\top \Sigma v \geq 0$ for all $v \in \mathbb{R}^n$. If X is not supported on a proper affine subspace, then $\Sigma \succ 0$.

Proof. Symmetry: $\Sigma_{ij} = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = \Sigma_{ji}$. PSD: $v^\top \Sigma v = \mathbb{E}[(v^\top (X - \mu))^2] \geq 0$. $\square$

When can $v^\top \Sigma v = 0$?

$$v^\top \Sigma v = 0 \implies \mathbb{E}[(v^\top (X - \mu))^2] = 0 \implies v^\top (X - \mu) = 0 \text{ a.s.}$$

The support of X lies within $\mu + \ker(v^\top)$, a hyperplane of dimension $n - 1$.

Remark. $\dim(\ker(v^\top)) = n - 1$ by rank-nullity: $\dim(\ker(v^\top)) + \text{rank}(v^\top) = n$, and $\text{rank}(v^\top) = 1$ since $v \neq 0$.

Remark (Support). The **support** of X is $\text{supp}(X) = \overline{\{x : p(x) > 0\}}$. " X is supported on S " means $P(X \in S) = 1$.

PROPERTIES OF COVARIANCE – SINGULARITY EXAMPLE

Example. Let $X = (X_1, X_2)$ with $X_2 = 2X_1$ always. Take $v = (2, -1)^\top$:

$$v^\top (X - \mu) = 2(X_1 - \mu_1) - (X_2 - \mu_2) = 2(X_1 - \mu_1) - 2(X_1 - \mu_1) = 0 \text{ a.s.}$$

so $v^\top \Sigma v = 0$. The direction normal to the constraint has zero variance – Σ is only PSD, not PD.

The support lies on the line $x_2 = 2x_1$ – a proper affine subspace of $\mathbb{R}^2$, confirming the proposition.

WHAT THE COVARIANCE MATRIX ENCODES

ENTRY	MEANING
Diagonal Σ_{ii}	Variance of component i — spread along axis i
Off-diagonal $\Sigma_{ij} > 0$	When X_i is above its mean, X_j tends to be above its mean too
Off-diagonal $\Sigma_{ij} < 0$	When X_i is above its mean, X_j tends to be below its mean
Off-diagonal $\Sigma_{ij} = 0$	No linear relationship between X_i and X_j

Robot example. For state $X = (p_x, p_y)$:

- Diagonal $\Sigma \rightarrow$ positional uncertainties in x and y are independent
- $\Sigma_{xy} > 0 \rightarrow$ corridor-like motion: overestimating p_x implies overestimating p_y

The Kalman filter tracks the full Σ because off-diagonal correlations carry information — collapsing to a diagonal approximation loses it.

CONDITIONAL INDEPENDENCE

Definition (Independence). $X \perp Y$ iff $p(x, y) = p(x)p(y)$ for all x, y — equivalently, $p(x | y) = p(x)$.

Definition (Conditional independence). $X \perp Y | Z$ iff $p(x, y | z) = p(x | z)p(y | z)$ for all x, y, z — equivalently, $p(x | y, z) = p(x | z)$.

Neither implies the other:

- $X \perp Y$ does **not** imply $X \perp Y | Z$. *Example:* X, Y are independent fair coin flips; $Z = \mathbf{1}[X = Y]$. Unconditionally $X \perp Y$, but given $Z = 1$ ("they match") and $X = H$, you know $Y = H$ — observing Z introduced a dependency.
- $X \perp Y | Z$ does **not** imply $X \perp Y$. *Example:* Z = rained, X = ground is wet, Y = umbrella sales. Given Z , X and Y are independent (rain explains both). But marginally, wet ground and umbrella sales are correlated — the shared cause Z creates marginal dependency.

The second case is the course structure: given x_t , past and future are conditionally independent — but marginally they are correlated through the state sequence.

THE MARKOV PROPERTY

Definition (Markov property). A stochastic process $(x_t)_{t \geq 0}$ satisfies the **Markov property** if:

$$x_t \perp x_{0:t-2} \mid x_{t-1} \quad \text{for all } t \geq 2$$

($x_{0:t-2}$ is empty for $t < 2$; the property holds vacuously.) The future depends on the past only through the present.

The state is a sufficient summary of history. If the Markov property fails in practice, the state definition is incomplete.

Definition (Observation Markov assumption). The observation model satisfies:

$$o_t \perp (o_{1:t-1}, a_{0:t-1}) \mid x_t \quad \implies \quad p(o_t \mid x_t, o_{1:t-1}, a_{0:t-1}) = p(o_t \mid x_t)$$

TRAJECTORY FACTORISATION

Proposition (Trajectory factorisation). For a controlled Markov system with policy π , the probability of a trajectory $\tau = (x_0, a_0, o_0, x_1, a_1, o_1, \dots, x_T)$ factorises as:

$$p_{\pi}(\tau) = p(x_0) \prod_{t=0}^{T-1} p(x_{t+1} \mid x_t, a_t) p(o_t \mid x_t) \pi(a_t \mid o_t)$$

Proof. Apply the chain rule repeatedly; use (i) the Markov property of dynamics and (ii) the Markov observation assumption to drop all history conditioning at each factor. $\square$

This factorisation is the starting point for every algorithm in the course: Kalman filter, particle filter, policy gradient, and the ELBO for latent dynamics models all follow from it.

THE GAUSSIAN DISTRIBUTION

Definition (Multivariate Gaussian). A random vector $X \in \mathbb{R}^n$ has distribution $X \sim \mathcal{N}(\mu, \Sigma)$ with mean $\mu \in \mathbb{R}^n$ and covariance $\Sigma \succ 0$ if its density is:

$$p(x) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right).$$

WHY THE GAUSSIAN DOMINATES

REASON	CONTENT
1. CLT	Physical noise = sum of many small errors → Gaussian by CLT
2. Max entropy	Highest entropy given fixed mean and variance — least informative prior
3. Conjugacy	Gaussian prior × Gaussian likelihood = Gaussian posterior (closed form)
4. Compact	Fully described by (μ, Σ) — $O(n^2)$ numbers, constant memory
5. Closure	Linear operations preserve Gaussianity — belief never leaves the family

These five properties compound: CLT justifies Gaussian noise, max entropy justifies Gaussian priors, conjugacy makes updates exact, compact parametrisation makes them tractable, closure makes them permanent.

ENTROPY

Definition (Shannon entropy). The **entropy** of X with distribution p is:

$$H(X) = -\mathbb{E}_p[\log p(X)]$$

Discrete: $H(X) = -\sum_x p(x) \log p(x)$. **Continuous (differential entropy):** $H(X) = -\int p(x) \log p(x) dx$.

Surprise interpretation. Each outcome x contributes $-\log p(x)$ surprise — rare events are surprising, certain events contribute zero. $H(X)$ is the *expected surprise* under p .

- High entropy $\rightarrow$ high uncertainty, many roughly equally likely outcomes
- Low entropy $\rightarrow$ concentrated distribution, outcome is nearly predictable
- $H = 0 \rightarrow$ deterministic: one outcome has probability 1

ENTROPY – SOURCE CODING

Bits (discrete, $\log_2$). By Shannon's source coding theorem, $H(X)$ equals the **minimum average number of bits** to encode a draw from p . The optimal code assigns length $-\log_2 p(x)$ to outcome x – short codes for likely outcomes, long codes for rare ones.

EXAMPLE	H
Certain outcome	0 bits
Fair coin $\text{Ber}(\frac{1}{2})$	1 bit
Fair die	$\log_2 6 \approx 2.58$ bits

ENTROPY – PROPERTIES

DISTRIBUTION	H
Bernoulli(θ)	$-\theta \log \theta - (1-\theta) \log(1-\theta)$; max at $\theta = \frac{1}{2}$
Uniform on $\{1, \dots, n\}$	$\log n$
$\mathcal{N}(\mu, \sigma^2)$	$\frac{1}{2} \ln(2\pi e \sigma^2)$ – depends on σ^2 only, not μ

Shift invariance. Location does not affect entropy – only spread does. Moving the Gaussian changes μ , not σ^2 , so H is unchanged.

Negative entropy (continuous). H can be negative for continuous distributions – e.g. $\mathcal{N}(0, \sigma^2)$ with small σ . This is not "negative information": encoding a real number to precision ε costs $H(X) - \log \varepsilon$ bits, which diverges as $\varepsilon \rightarrow 0$ for every distribution. Only *differences* $H(p) - H(q)$ are meaningful. Negative H means the distribution is more concentrated than $\text{Uniform}[0, 1]$ (which has $H = 0$).

KL DIVERGENCE

Definition (KL divergence). The **Kullback–Leibler divergence** from p to q is:

$$D_{\text{KL}}(p\|q) = \mathbb{E}_p \left[\log \frac{p(X)}{q(X)} \right] = \int p(x) \log \frac{p(x)}{q(x)} dx$$

How much does using q as a model cost, when the truth is p ?

Connection: $D_{\text{KL}}(p\|q) = H(p, q) - H(p)$ where $H(p, q) = -\mathbb{E}_p[\log q(X)]$ is the **cross-entropy**. Minimising KL $\equiv$ minimising cross-entropy (the classification loss).

Proposition (Gibbs' inequality). $D_{\text{KL}}(p\|q) \geq 0$, with equality iff $p = q$ a.e.

Proof. Jensen on $-\log$: $\mathbb{E}_p \left[-\log \frac{q}{p} \right] \geq -\log \mathbb{E}_p \left[\frac{q}{p} \right] = -\log \int q dx = 0. \square$

You can never do better (in expected coding cost) by using the wrong model.

KL DIVERGENCE – FORWARD VS REVERSE

Forward KL $\min_q D_{\text{KL}}(p||q)$ – **mean-seeking / inclusive**.

If $p(x) > 0$ but $q(x) = 0$, the integrand $p(x) \log \frac{p(x)}{q(x)} = +\infty$. So q must place mass everywhere p does. For a Gaussian q , the minimiser matches the moments: $\mu_q = \mathbb{E}_p[X]$, $\Sigma_q = \text{Cov}_p(X)$. Covers all modes but may fill low-probability valleys.

Reverse KL $\min_q D_{\text{KL}}(q||p)$ – **mode-seeking / exclusive**.

If $q(x) > 0$ but $p(x) \approx 0$, the integrand $\rightarrow +\infty$. So q avoids regions where p is small – collapses onto a single mode, ignores the rest.

Example – bimodal p . $p = 0.5 \mathcal{N}(-2, 0.25) + 0.5 \mathcal{N}(2, 0.25)$, $q = \mathcal{N}(\mu, \sigma^2)$:

- Forward KL: $\mu \approx 0, \sigma \approx 2$ – straddles both modes, wrong everywhere.
- Reverse KL: $\mu \approx \pm 2, \sigma \approx 0.5$ – accurate near one mode, ignores the other.

§3 · BAYES' THEOREM

UPDATING BELIEF FROM EVIDENCE



BAYES' THEOREM

Theorem (Bayes). For random variables X and Y with $p(y) > 0$:

$$p(x | y) = \frac{p(y | x) p(x)}{p(y)}, \quad p(y) = \int p(y | x) p(x) dx$$

Proof. Chain rule: $p(x, y) = p(x | y)p(y) = p(y | x)p(x)$. Divide by $p(y)$. $\square$

TERM	SYMBOL	ROLE
Prior	$p(x)$	Belief about x before seeing o
Likelihood	$p(o x)$	Probability of observing o if state is x
Posterior	$p(x o)$	Updated belief after observing o
Marginal likelihood	$p(o)$	Normalising constant — ensures $\int p(x o) dx = 1$

THE POSTERIOR IS THE BELIEF

$$b_t = p(x_t \mid o_{1:t})$$

Sequential Bayes: each posterior becomes the prior for the next observation.

$$p(x \mid o_1, o_2) \propto p(o_2 \mid x) \underbrace{p(x \mid o_1)}_{\text{yesterday's posterior} = \text{today's prior}}$$

Remark (Bayesian vs frequentist). The frequentist treats x as a fixed unknown parameter; estimation produces a point estimate and a confidence interval (a property of the sampling procedure). The Bayesian treats x as a random variable; $b_t = p(x_t \mid o_{1:t})$ is the complete answer to "what do I believe about the state?" — not because of a philosophical commitment, but because maintaining a full distribution is the correct solution to partial observability.

S4 · OPTIMISATION RECAP

FINDING THE BEST POLICY



THE OPTIMISATION PROBLEM

$$\min_{\theta \in \Theta} \mathcal{L}(\theta) \quad \text{s.t.} \quad h_i(\theta) = 0, \quad i = 1, \dots, m \quad \text{and} \quad g_j(\theta) \leq 0, \quad j = 1, \dots, r$$

Definition (Local minimum). θ^* is a **local minimum** if $\exists \varepsilon > 0$ such that $\mathcal{L}(\theta^*) \leq \mathcal{L}(\theta)$ for all $\theta \in \Theta$ with $\|\theta - \theta^*\| < \varepsilon$.

Definition (Global minimum). θ^* is a **global minimum** if $\mathcal{L}(\theta^*) \leq \mathcal{L}(\theta)$ for all $\theta \in \Theta$.

TERM	MEANING
$\mathcal{L}(\theta)$	Objective (loss) to minimise
$h_i(\theta) = 0$	Equality constraints
$g_j(\theta) \leq 0$	Inequality constraints
$m = r = 0$	Unconstrained problem

CONVEXITY – DEFINITIONS

Definition (Convex set). $\Theta \subseteq \mathbb{R}^n$ is **convex** if for all $\theta, \phi \in \Theta$ and $\lambda \in [0, 1]$:

$$\lambda\theta + (1 - \lambda)\phi \in \Theta$$

Definition (Convex function). $\mathcal{L}$ is **convex** on a convex Θ if for all $\theta, \phi \in \Theta, \lambda \in [0, 1]$:

$$\mathcal{L}(\lambda\theta + (1 - \lambda)\phi) \leq \lambda \mathcal{L}(\theta) + (1 - \lambda) \mathcal{L}(\phi)$$

Equivalently (differentiable case): $\mathcal{L}(\phi) \geq \mathcal{L}(\theta) + \nabla \mathcal{L}(\theta)^\top (\phi - \theta)$ – $\mathcal{L}$ lies above every tangent hyperplane.

Theorem (Global optimality for convex objectives). If $\mathcal{L}$ is convex on a convex set Θ , then every local minimum is a **global minimum**.

Proof. Let θ^* be a local min and $\phi \in \Theta$ arbitrary. Convexity: $\mathcal{L}(\lambda\phi + (1-\lambda)\theta^*) \leq \lambda\mathcal{L}(\phi) + (1-\lambda)\mathcal{L}(\theta^*)$. For small $\lambda > 0$, local minimality gives $\mathcal{L}(\theta^*) \leq \mathcal{L}(\phi)$. $\square$

STRONG CONVEXITY

Definition (Strongly convex). $\mathcal{L}$ is μ -strongly convex ($\mu > 0$) if for all θ, ϕ :

$$\mathcal{L}(\phi) \geq \mathcal{L}(\theta) + \nabla \mathcal{L}(\theta)^\top (\phi - \theta) + \frac{\mu}{2} \|\phi - \theta\|^2$$

Equivalently: $\nabla^2 \mathcal{L}(\theta) \succeq \mu I$ at every θ .

Proposition (Unique global minimum). If $\mathcal{L}$ is μ -strongly convex, it has a **unique global minimum**.

Proof. Suppose $\theta^* \neq \phi$ are both global minima. Strong convexity gives $\mathcal{L}(\theta^*) \geq \mathcal{L}(\phi) + \frac{\mu}{2} \|\theta^* - \phi\|^2 > \mathcal{L}(\phi)$, contradicting minimality of θ^* . $\square$

Strong convexity means $\mathcal{L}$ curves up at least as fast as a quadratic with curvature μ — no flat regions of minima.

GRADIENTS AND FIRST-ORDER CONDITIONS

Definition (Gradient). For differentiable $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$:

$$\nabla_{\theta} \mathcal{L}(\theta) = \left(\frac{\partial \mathcal{L}}{\partial \theta_1}, \dots, \frac{\partial \mathcal{L}}{\partial \theta_n} \right)^{\top} \in \mathbb{R}^n$$

The gradient points in the direction of **steepest ascent**.

Definition (Hessian). For twice-differentiable $\mathcal{L}$: $\nabla_{\theta}^2 \mathcal{L}(\theta) = \left(\frac{\partial^2 \mathcal{L}}{\partial \theta_i \partial \theta_j} \right)_{i,j} \in \mathbb{R}^{n \times n}$. The Hessian is symmetric (Clairaut's theorem).

Theorem (First-order necessary condition). If θ^* is a local minimum of differentiable $\mathcal{L}$ on an open set, then:

$$\nabla_{\theta} \mathcal{L}(\theta^*) = 0$$

Proof. If $\nabla \mathcal{L}(\theta^*) \neq 0$, then $\theta^* - \varepsilon \nabla \mathcal{L}(\theta^*)$ has strictly smaller value for small $\varepsilon > 0$. $\square$

SECOND-ORDER CLASSIFICATION

Theorem (Second-order classification). Let θ^* be a stationary point with Hessian $H^* = \nabla^2 \mathcal{L}(\theta^*)$. Then:

CONDITION ON H^*	CONCLUSION
$H^* \succ 0$ (positive definite)	θ^* is a strict local minimum
$H^* \prec 0$ (negative definite)	θ^* is a strict local maximum
H^* indefinite (mixed eigenvalues)	θ^* is a saddle point
$H^* \succeq 0$ or $H^* \preceq 0$ (semidefinite)	inconclusive — higher-order terms decide

Proof sketch. Taylor-expand: $\mathcal{L}(\theta^* + d) = \mathcal{L}(\theta^*) + \frac{1}{2}d^\top H^* d + O(\|d\|^3)$. The sign of $d^\top H^* d$ for all $d \neq 0$ determines the case. $\square$

GRADIENT DESCENT AND SADDLES

Remark (Gradient descent). The standard iterative algorithm for unconstrained minimisation:

$$\theta_{k+1} = \theta_k - \alpha \nabla_{\theta} \mathcal{L}(\theta_k), \quad \alpha > 0$$

Convergence: $O(1/k)$ for convex $\mathcal{L}$ with step $\alpha \leq 1/L$; linear rate $O((1 - \mu/L)^k)$ for μ -strongly convex.

Saddle points in non-convex optimisation:

- Far more common than local minima in neural network training and policy search
- Gradient near a saddle stalls ($\nabla \mathcal{L} \approx 0$) but the Hessian has descent directions
- SGD noise and adaptive methods (Adam, natural gradient) help escape

For convex objectives: every stationary point is a global minimum — saddles do not exist.

CONSTRAINED OPTIMISATION – LAGRANGIAN

Definition (Lagrangian). For the problem with equality constraints $h_i(\theta) = 0$ and inequality constraints $g_j(\theta) \leq 0$:

$$L(\theta, \lambda, \mu) = \mathcal{L}(\theta) + \sum_{i=1}^m \lambda_i h_i(\theta) + \sum_{j=1}^r \mu_j g_j(\theta)$$

$\lambda \in \mathbb{R}^m$ – **equality multipliers**; $\mu \in \mathbb{R}_{\geq 0}^r$ – **inequality multipliers** (KKT multipliers).

Remark (Naming). KKT stands for **Karush–Kuhn–Tucker**. William Karush derived the conditions in his 1939 master's thesis. Harold Kuhn and Albert Tucker independently rediscovered and published them in 1951 – the name reflects the belated recognition of Karush's priority.

CONSTRAINT QUALIFICATION

Definition (Constraint qualification). A regularity condition at θ^* ensuring that KKT multipliers exist and are finite.

Why is it needed? KKT multipliers are solutions to the stationarity equation $\nabla \mathcal{L} + \lambda^\top \nabla h + \mu^\top \nabla g = 0$. If the gradients of active constraints are linearly dependent at θ^* , this system becomes degenerate — multipliers can blow up or fail to exist entirely.

An **active constraint** at θ^* is one satisfied with equality: $g_j(\theta^*) = 0$. Inactive constraints ($g_j(\theta^*) < 0$) don't affect the local geometry and are ignored.

CONSTRAINT QUALIFICATION – LICQ

Sufficient condition – LICQ (Linear Independence CQ): gradients of all active constraints

$$\{\nabla h_i(\theta^*)\} \cup \{\nabla g_j(\theta^*) : g_j(\theta^*) = 0\}$$

are linearly independent at θ^* . This is the generic condition — it fails only in degenerate, measure-zero configurations (e.g. two constraints exactly tangent to each other).

Remark. In practice, CQs are rarely checked explicitly — LICQ holds generically. The exception is when a solver returns very large or inconsistent multipliers, which signals a CQ violation at the solution.

KKT NECESSARY CONDITIONS

Theorem (KKT necessary conditions). Let θ^* be a local minimum with a constraint qualification holding. Then $\exists \lambda^* \in \mathbb{R}^m, \mu^* \in \mathbb{R}^r$ such that:

1. **Stationarity:** $\nabla_{\theta} \mathcal{L}(\theta^*) + \sum_i \lambda_i^* \nabla h_i(\theta^*) + \sum_j \mu_j^* \nabla g_j(\theta^*) = 0$
2. **Primal feasibility:** $h_i(\theta^*) = 0$ for all i ; $g_j(\theta^*) \leq 0$ for all j
3. **Dual feasibility:** $\mu_j^* \geq 0$ for all j
4. **Complementary slackness:** $\mu_j^* g_j(\theta^*) = 0$ for all j

Corollary (Sufficiency for convex programs). If $\mathcal{L}$ and all g_j are convex, all h_i are affine, and $(\theta^*, \lambda^*, \mu^*)$ satisfy the KKT conditions, then θ^* is a **global minimum**.

KKT – INTUITION

Complementary slackness: either the constraint is *inactive* ($g_j < 0 \Rightarrow \mu_j^* = 0$, plays no role) or *active* ($g_j = 0 \Rightarrow \mu_j^* \geq 0$, pushes back against the objective).

Why $\mu_j^* \geq 0$? The constraint gradient ∇g_j points outward (toward infeasibility). Stationarity $\nabla \mathcal{L} = -\sum_j \mu_j^* \nabla g_j$ requires non-negative μ_j^* to balance an inward-pointing objective gradient. Geometrically: the constraint is a wall; the multiplier is the normal force — always outward, never pulling.

KKT IN THIS COURSE

LECTURE	WHERE KKT APPEARS
Lec 8 — HJB & PMP	Costate $\lambda(t)$ is the KKT multiplier for the dynamics constraint $\dot{x} = f(x, a)$
Lec 8 — LQR	The Riccati equation is the KKT stationarity condition for the finite-horizon linear-quadratic Lagrangian
Lec 16 — Constrained policy opt.	Safety constraints $g(\pi) \leq 0$ handled by augmenting the policy loss with a KKT multiplier; dual variable tracks constraint violation

Remark (Shadow price). The multiplier λ_i^* is the **shadow price** of constraint i : it measures how much the optimal objective changes if the constraint is relaxed from $h_i(\theta) = 0$ to $h_i(\theta) = \varepsilon$, to first order: $d\mathcal{L}^*/d\varepsilon|_0 = -\lambda_i^*$.

S5 · MAP ESTIMATION

WHERE INFERENCE MEETS OPTIMISATION



THE MAP ESTIMATOR

Definition (MAP estimator). The **maximum a posteriori** estimator is the mode of the posterior:

$$\hat{x}_{\text{MAP}} = \arg \max_x p(x \mid o) = \arg \max_x [\log p(o \mid x) + \log p(x)]$$

Equivalently:

$$\hat{x}_{\text{MAP}} = \arg \min_x [-\log p(o \mid x) - \log p(x)]$$

Key insight: MAP converts a Bayesian inference problem into an optimisation problem — minimise negative log-posterior.

INFERENCE VIEW	OPTIMISATION VIEW
Posterior $p(x \mid o)$	Objective $-\log p(o \mid x) - \log p(x)$
Normalising constant $p(o)$	Additive constant (dropped)
Mode of the posterior	Stationary point of the objective

MAP AS REGULARISED LEAST SQUARES

Proposition (MAP as regularised least squares). With prior $x \sim \mathcal{N}(0, \Sigma_0)$ and likelihood $o \mid x \sim \mathcal{N}(Hx, R)$, the MAP estimator solves:

$$\hat{x}_{\text{MAP}} = \arg \min_x \underbrace{(o - Hx)^\top R^{-1} (o - Hx)}_{\text{data fit}} + \underbrace{x^\top \Sigma_0^{-1} x}_{\text{regulariser}},$$

where Σ_0^{-1} is a regularisation matrix.

Proof. $-\log p(o \mid x) - \log p(x) = \frac{1}{2}(o - Hx)^\top R^{-1}(o - Hx) + \frac{1}{2}x^\top \Sigma_0^{-1}x + \text{const.} \quad \square$

Corollary (Closed form). Setting $\nabla_x[\dots] = 0$:

$$\hat{x}_{\text{MAP}} = (\Sigma_0^{-1} + H^\top R^{-1} H)^{-1} H^\top R^{-1} o$$

Tight prior (small Σ_0) $\rightarrow$ strong regularisation; diffuse prior ($\Sigma_0 \rightarrow \infty$) $\rightarrow$ ordinary least squares.

MAP EQUALS POSTERIOR MEAN

Proposition (MAP equals posterior mean for Gaussian models). Under the Gaussian prior and Gaussian likelihood of the preceding Proposition, the MAP estimator coincides with the posterior mean:

$$\hat{x}_{\text{MAP}} = \mathbb{E}[x \mid o]$$

For non-Gaussian or multimodal posteriors, $\text{MAP} \neq \text{posterior mean}$ — MAP returns only the dominant mode.

MAP IN PRACTICE

ALGORITHM	MAP FORM
Ridge regression	MAP with Gaussian prior (ℓ_2 penalty)
Kalman smoother	MAP over full trajectory $x_{0:T}$ given all $o_{0:T}$
Bundle adjustment (SLAM)	MAP over camera poses and 3D landmarks
Neural nets with weight decay	MAP with Gaussian prior on weights; $\lambda = 1/\sigma_0^2$
MLE	MAP with uniform (improper) prior — likelihood only

When to use MAP vs. full posterior:

- **MAP** — single answer needed, posterior is unimodal; cheaper than full posterior
- **Full posterior** — uncertainty matters downstream (e.g. Kalman filter needs Σ_t to compute the gain at $t + 1$; discarding it breaks sequential estimation)

S6 · THE DUALITY

INFERENCE AND OPTIMISATION ARE TWO VIEWS OF THE SAME OBJECT



THE ALGORITHM MAP

Known model	Kalman filter, Particle filter	LQR, MPC, A*
Learned / unknown	Bayesian state estimation	Behavioural cloning, IRL
Both	—	RL (policy gradient + value estimation) · Diffusion policy

The top-left and bottom-right corners are "pure" — single-language algorithms. RL occupies the hard corner: it must optimise π while simultaneously *inferring* value estimates from noisy interaction.

RL is hardest because it lives in both languages at once.

INFERENCE AND OPTIMISATION – TWO VIEWS OF ONE OBJECT

The unifying object is the **trajectory distribution**:

$$p_{\pi}(\tau) = p(x_0) \prod_t p(x_{t+1} \mid x_t, a_t) p(o_t \mid x_t) \pi(a_t \mid o_t)$$

- **Inference** computes *marginals* of $p_{\pi}(\tau)$ – what is $p(x_t \mid o_{1:t})$?
- **Optimisation** *extremises* a functional of $p_{\pi}(\tau)$ – what is $\arg \min_{\pi} \mathcal{L}(\pi)$?

The language shift — from marginals to extrema — is what distinguishes the two algorithm families.

WHAT'S NEXT

Lecture 5 — State Estimation

- Inferring x_t from observations $o_{1:t}$ — the hidden-state problem
- Belief state $b_t = p(x_t \mid o_{1:t})$ as a sufficient statistic for decisions
- Trajectory factorisation from today → Bayesian belief update
- Kalman filter: exact closed-form inference when f and h are linear Gaussian
- Particle filter: nonlinear, non-Gaussian case

Every probability tool from today is used in the very next lecture.